

David Matos

Software / Backend Engineer

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SUMMARY

Computer Science graduate with internship experience in **Python backend development**, **data processing**, **reusable software**, **CI/CD**, and **cloud-native technologies**. Former **professional handball player** with **10 years of competitive team experience**.

EDUCATION

Faculty of Sciences, University of Porto

Porto, Portugal

B.Sc. in Computer Science

2022 – 2026

- Relevant coursework: Software Engineering, Distributed Systems, Algorithms, Databases, Large-Scale Data Science, Machine Learning, and Problem Solving.

EXPERIENCE

Summer Intern – Software Engineering

Summer 2025

Celfinet, a Cyient Company

Lisbon, Portugal

- Refactored and restructured an existing **FastAPI backend** used for base-station energy consumption studies, improving code organization, maintainability, and component reusability.
- Developed and maintained reusable **Python components across dozens of internal packages** for processing **very large datasets**, improving code reuse and maintainability across data-intensive workflows. The developed packages were actively adopted and used by an internal team.
- Developed a **CSV data validation and error-reporting system** to identify missing values, invalid data, incorrect units, and duplicate records in customer-submitted files.
- Implemented automated correction mechanisms for selected data-quality issues, including **missing values**, **invalid data**, **unit inconsistencies**, and **duplicates**, where reliable correction rules could be applied.
- Developed and maintained **automated tests** to validate backend functionality and data-processing behavior.

Curricular Intern – Software Engineering

October 2025 – January 2026

Celfinet, a Cyient Company

Lisbon, Portugal

- Designed and developed a reusable **Python library** with hundreds of functions for direct access to frequently used datasets, simplifying and accelerating data extraction for ad-hoc studies.
- Built the library to interact directly with internal databases using **SQLAlchemy**, providing reusable abstractions that reduced the complexity of recurring data-access workflows.
- Implemented comprehensive **unit tests** to validate library functionality and ensure reliability across its data-access components.
- Automated the library's testing, packaging, and internal distribution through a **CI/CD pipeline** in the company's Azure DevOps repository, enabling developers to install and use the library directly.
- Adopted by **two internal teams** and still in use after the completion of the internship, demonstrating continued adoption and practical value within the organization.

Vice President

2023 – 2024

NUCC – Student Nucleus of Computer Science

Faculty of Sciences, University of Porto

- Organized technical and career-oriented events connecting Computer Science students with industry professionals and external organizations.
- Coordinated communication and execution across **students**, **faculty members**, **industry professionals**, and **external organizations**.
- Managed multiple initiatives within the Computer Science student community, developing practical experience in coordination, communication, and leadership.

PROJECTS

- Personal Cloud Platform for Large-Scale Data Science** | *Python, Docker, Kubernetes, Linux* 2026
- Designed and implemented a **cloud-based platform for large-scale data science workloads**, using containerized services and cloud-native technologies.
 - Used **Docker and Kubernetes** to package, deploy, and orchestrate application components, exploring container orchestration and scalable system deployment.
 - Designed the system with a focus on **modularity, scalability, and automated deployment** within a Linux-based environment.
- K-Nearest Neighbors (KNN) Model Optimization** | *Python, Machine Learning, scikit-learn* 2026
- Implemented and evaluated the **K-Nearest Neighbors (KNN)** algorithm in Python, analyzing model performance across different datasets and configurations.
 - Applied **K-Fold cross-validation** to evaluate model generalization and compare different KNN configurations using consistent evaluation procedures.
 - Investigated the impact of **feature scaling, PCA dimensionality reduction**, and **Manhattan distance** on model performance and classification results.
 - Compared different preprocessing and model configurations, analyzing their impact on **accuracy, generalization, and computational performance**.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript
Backend: FastAPI, Flask, Django, REST APIs, SQLAlchemy
Cloud & DevOps: Docker, Kubernetes, CI/CD, Linux, Bash, Azure, GCP
Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Automated Testing, Git, Version Control
Data & Machine Learning: Data Processing, Data Analysis, Feature Engineering, Machine Learning Fundamentals, Model Evaluation
Tools: Postman
Languages: Portuguese (Native), English (Professional)

ADDITIONAL EXPERIENCE

Professional Handball Player 2012 – 2022
Competitive Team Sport *Portugal*

- * Competed at a high level for 10 years, developing strong experience in teamwork, discipline, communication, and performance under pressure.